Exploratory Data Analysis Report

Plant Disease Detection Dataset

Submitted by

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1. Introduction

This report presents the results of an exploration data analysis (EDA) conducted on a plant disease detection dataset. The primary objective of this analysis was to understand the structure of the data, examine the statistical properties of each variable, identify potential issues with data quality, and draw preliminary conclusions about the relationship between environmental conditions and the occurrence of plant disease. These insights are intended to guide the feature engineering and modelling decisions that will follow in subsequent stages of the project.

The dataset consists of 10,000 observations collected from agricultural settings, capturing four environmental measurements alongside a binary indicator of whether a plant was found to be diseased. The analysis was carried out using Python, with the panda's library used for data manipulation and matplotlib and seaborn used for visualisation.

2. Dataset Overview

The dataset contains 10,000 rows and five columns. Four columns represent continuous environmental features recorded at the time of observation: temperature (in degrees Celsius), relative humidity (as a percentage), rainfall (in millimetres), and soil pH. The fifth column, disease_present, serves as the target variable and takes the value 0 if the plant was healthy and 1 if it showed signs of disease.

An initial inspection of the dataset confirmed that there were no missing values in any column, which is an encouraging characteristic as it means no imputation strategy was required before proceeding with the analysis. The data types were also appropriate for the nature of each variable, with the four environmental features stored as 64-bit floating-point numbers and the binary target as a 64-bit integer.

Table 1: Structure and description of all columns in the dataset

Column Name	Data Type	Role in Analysis
temperature	float64	Input feature
humidity	float64	Input feature
rainfall	float64	Input feature
soil_pH	float64	Input feature
disease_present	int64	Target variable (binary: 0 or 1)

3. Class Distribution

Understanding the balance of the target variable is an essential first step in any classification task. An imbalanced dataset can lead a machine learning model to achieve high accuracy simply by predicting the majority class for every observation, which would be misleading and practically useless.

As shown in Table 2 and Figure 1 below, the dataset is notably imbalanced. Of the 10,000 records, 7,590 belong to the healthy class (class 0), representing 75.9% of the dataset, while only 2,410 records belong to the diseased class (class 1), representing the remaining 24.1%. This is a roughly three-to-one ratio in favour of healthy observations.

This imbalance is important to keep in mind throughout the rest of the analysis and must be explicitly addressed when training any predictive model. Techniques such as assigning class weights inversely proportional to class frequencies, oversampling the minority class using methods like SMOTE, or using evaluation metrics that account for class imbalance, such as the F1-score or the area under the ROC curve, will be necessary to ensure that the model learns to detect disease effectively rather than simply defaulting to predicting healthy.

Table 2: Distribution of the target variable across the two class labels

Class Label	Number of Records	Proportion of Dataset
0 (Healthy plant)	7,590	75.9%
1 (Diseased plant)	2,410	24.1%

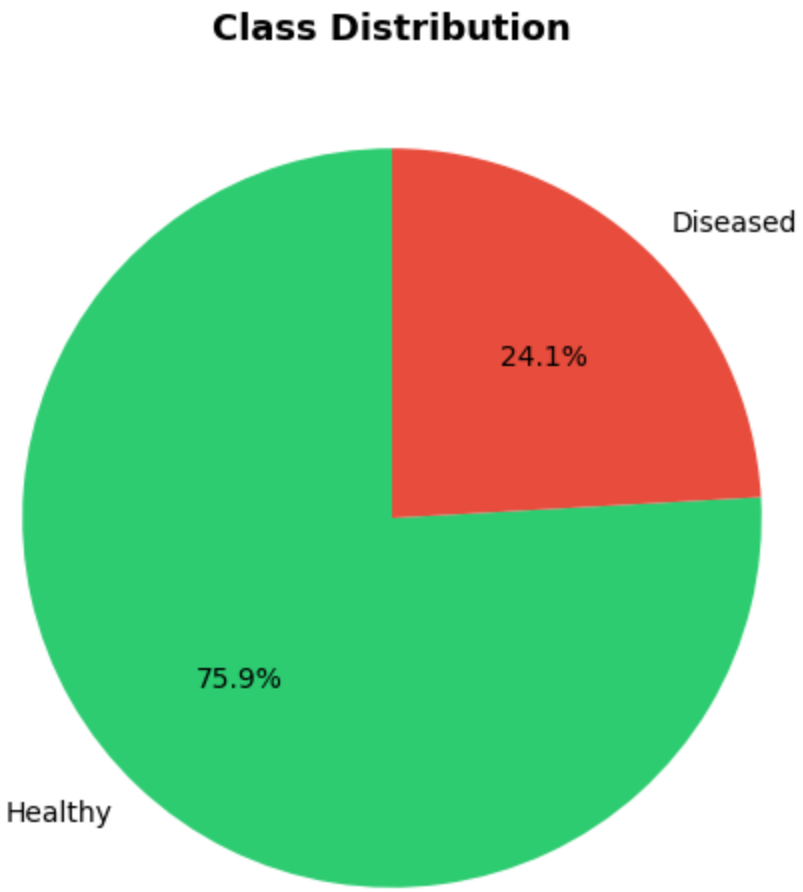

Figure 1: Class distribution of the target variable (disease_present)

4. Descriptive Statistics

Table 3 below presents a summary of the key descriptive statistics for each of the four environmental features in the dataset. These statistics offer a concise picture of the central tendency, spread, and range of each variable.

Table 3: Descriptive statistics for all four environmental features

Feature	Mean	Std. Dev.	Minimum	Maximum	IQR Outliers
Temperature (°C)	25.61	5.81	5.39	56.69	245
Humidity (%)	62.12	22.68	6.24	102.40	0
Rainfall (mm)	9.81	9.85	0.00	84.65	496

Feature	Mean	Std. Dev.	Minimum	Maximum	IQR Outliers
Soil pH	6.25	1.30	4.00	8.50	0

4.1 Temperature

Temperature values in the dataset range from a minimum of 5.39°C to a maximum of 56.69°C, with a mean of approximately 25.61°C and a standard deviation of 5.81°C. The mean is close to what would be expected for agricultural settings in many temperate and tropical regions. However, the maximum value of 56.69°C warrants attention, as this level of heat is above the physiological tolerance limit of most crop plants. Fifteen records in the dataset have temperatures exceeding 50°C, which suggests either sensor malfunction or a data entry error. The IQR method identified 245 records as statistical outliers for this feature.

4.2 Humidity

Humidity has a mean of 62.12% and a standard deviation of 22.68%, indicating a fairly widespread around the average. The minimum recorded value of 6.24% suggests very dry conditions, while the maximum of 102.40% is physically impossible, as relative humidity cannot exceed 100%. One record in the dataset contains this erroneous value and should be corrected or removed prior to modelling. Despite this anomaly, the IQR method found no outliers for this feature, which reflects the fact that the extreme value falls just slightly above the physically valid range rather than being statistically distant from the rest of the distribution.

4.3 Rainfall

Among the four features, rainfall exhibits the most notable distributional asymmetry. The mean is 9.81 mm with a standard deviation of 9.85 mm, meaning the standard deviation is almost as large as the mean itself. The minimum value is essentially zero while the maximum reaches 84.65 mm, suggesting a long right tail caused by occasional heavy rainfall events. This kind of right-skewed distribution is common in meteorological data. The IQR method flagged 496 records as outliers, accounting for approximately five percent of the entire dataset. This skewness should be addressed through a log-transformation or Winsorisation before rainfall is used as a model input.

4.4 Soil pH

Soil pH is the most well-behaved of the four features. Its values span from 4.00 to 8.50, which falls entirely within the range of naturally occurring soil acidity and alkalinity. The mean of 6.25 and standard deviation of 1.30 are reasonable for agricultural land, and the IQR method identified no outliers for this variable. Soil pH can therefore be used in modelling without any special preprocessing beyond standard scaling.

5. Feature Distributions by Class

To understand which environmental conditions are associated with plant disease, we examined how each feature is distributed separately for healthy and diseased plants. This was done using overlapping histograms and side-by-side box plots, which together provide a comprehensive picture of both the shape of each distribution and the location of the central values and spread.

5.1 Histograms

Figure 2 shows the histogram for each feature, with the green bars representing healthy plants and the red bars representing diseased plants. The distributions for temperature are almost completely overlapping, with both classes following a roughly bell-shaped curve centred near the overall mean. This suggests that temperature, taken in isolation, is unlikely to be a strong predictor of disease status.

The histogram for humidity tells a different story. While both distributions share a similar shape, the distribution for diseased plants is shifted noticeably to the right, indicating that diseased plants tend to be associated with higher humidity levels. A similar, though slightly less pronounced, rightward shift can be observed for rainfall in the diseased class. For soil pH, the diseased class appears to be very slightly skewed toward lower, more acidic values compared to the healthy class, though the two distributions are broadly similar.

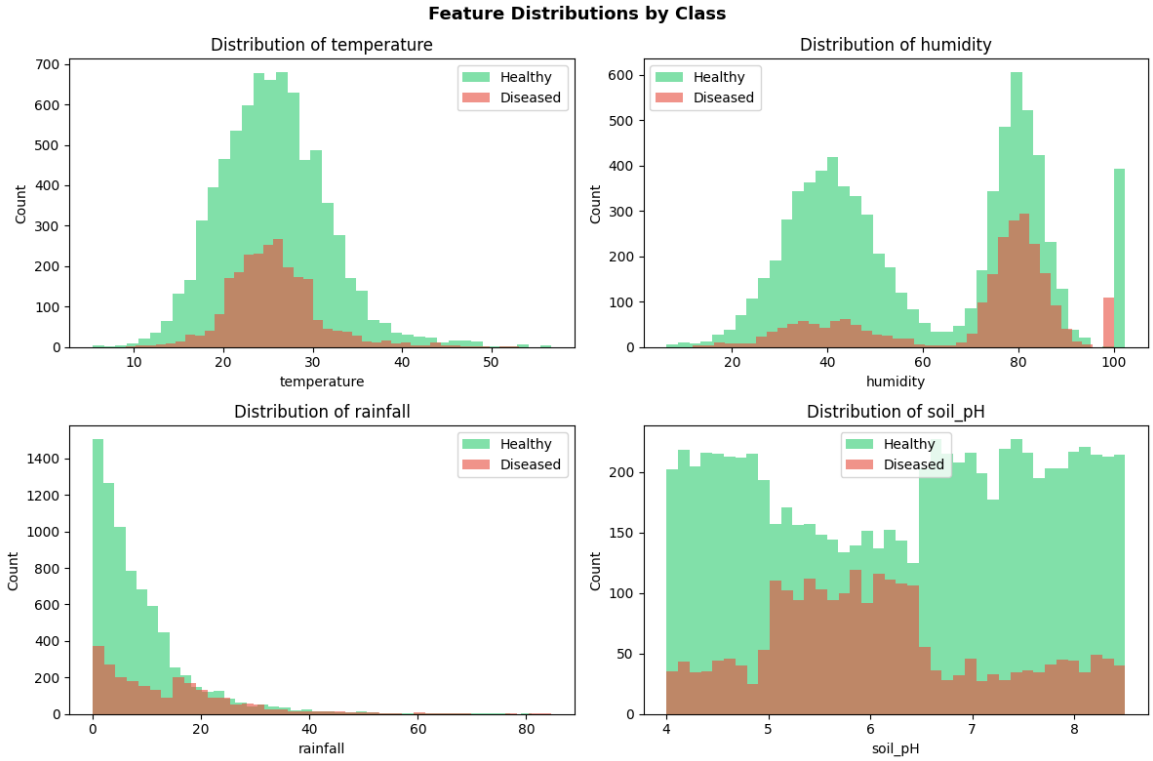

Figure 2: Distribution of each environmental feature separated by class

5.2 Box Plots

The box plots in Figure 3 reinforce and clarify the observations made from the histograms. For temperature, the median and interquartile range of both classes are virtually identical, confirming that temperature is not a meaningful discriminating feature on its own.

For humidity, the median for the diseased class is visibly higher than that for the healthy class, and the interquartile range is shifted upward. This is the most visually apparent difference between the two classes across all four features. Rainfall shows a similar pattern but to a lesser degree, with the median for diseased plants sitting somewhat above that for healthy plants. The soil pH box plot shows a slightly lower median for the diseased class, consistent with the earlier observation that disease tends to occur in slightly more acidic conditions.

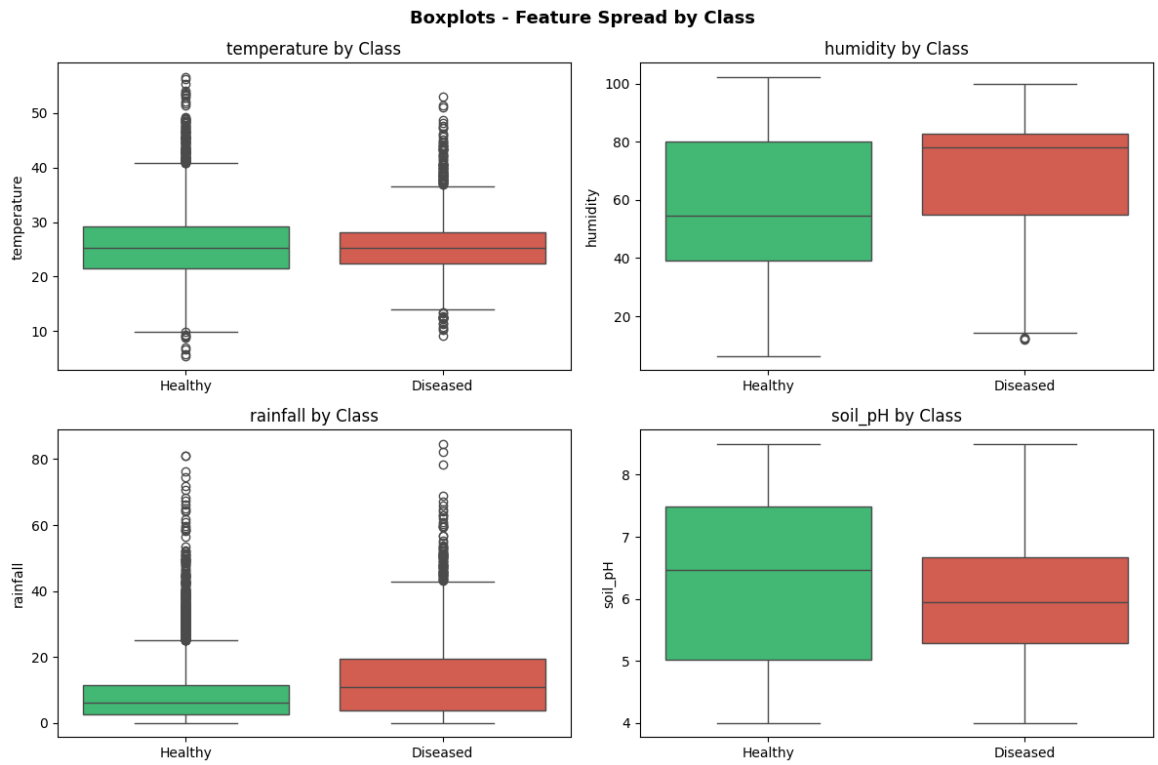

Figure 3: Box plots comparing feature spread across healthy and diseased classes

6. Mean Feature Values by Class

To make the differences between classes more concrete, Table 4 presents the mean value of each feature broken down by class. These figures confirm the patterns observed in the visualisations above.

Table 4: Mean values of each environmental feature for healthy and diseased plants

Class	Temperature (°C)	Humidity (%)	Rainfall (mm)	Soil pH
0 (Healthy)	25.615	59.573	8.594	6.300
1 (Diseased)	25.603	70.138	13.629	6.075

The most striking contrast between the two classes is in humidity, where diseased plants have an average humidity of 70.14% compared to 59.57% for healthy plants, a difference of more than ten percentage points. Rainfall also shows a notable gap, with diseased plants experiencing an average of 13.63 mm compared to 8.59 mm for healthy plants. Soil pH is modestly lower for

diseased plants at 6.08 versus 6.30 for healthy ones. Temperature, as expected, shows virtually no difference between the two classes, with both hovering around 25.6°C.

These differences confirm that humidity and rainfall are the most informative features in this dataset with respect to predicting plant disease, while temperature contributes little discriminative information on its own.

7. Correlation Analysis

A Pearson correlation analysis was carried out to quantify the linear relationships between all pairs of variables in the dataset. The resulting correlation matrix is displayed as a heatmap in Figure 4, where warmer colours indicate positive correlations and cooler colours indicate negative correlations.

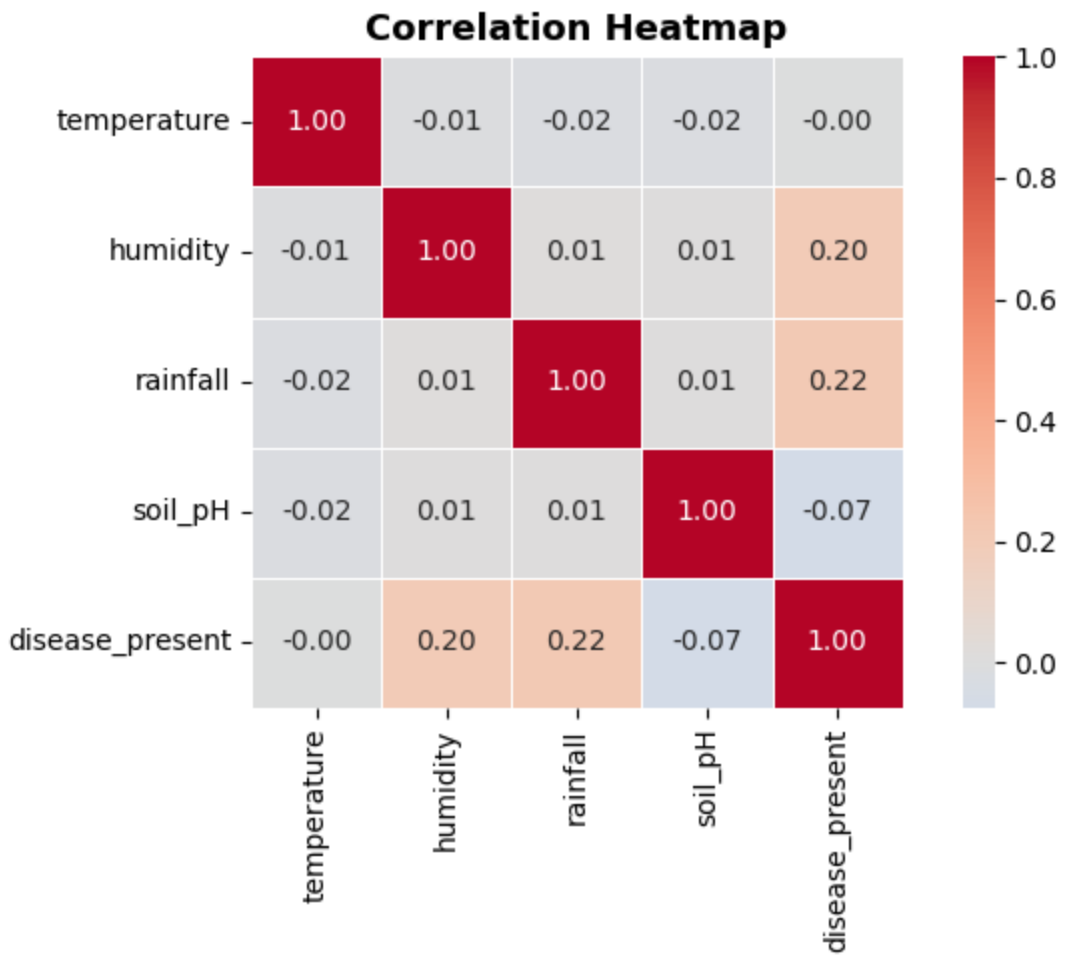

Figure 4: Pearson correlation heatmap across all variables

The correlations between the input features and the target variable align with what the distributional analysis suggested. Humidity shows the highest positive correlation with disease_present at approximately 0.20, meaning that as humidity increases, the likelihood of disease presence also increases. Rainfall follows with a moderate positive correlation of around 0.13. Soil pH shows a small negative correlation of roughly negative 0.08, meaning that lower pH values are marginally associated with higher disease incidence. Temperature has a correlation close to zero, confirming that it carries almost no linear relationship with disease occurrence.

Among the input features themselves, no strong correlations are present. The highest inter-feature correlation is between rainfall and humidity, which is expected given that wet conditions naturally produce both. However, even this correlation is not strong enough to raise concerns about multicollinearity in a linear modelling setting. This means all four features can reasonably be included together as inputs without one dominating or suppressing the effect of another.

8. Outlier Analysis and Data Quality

Identifying outliers and data quality issues is a critical part of any exploratory analysis. Outliers that arise from genuine extreme conditions may carry important information and should be retained, while those caused by sensor errors or data entry mistakes should be corrected or removed to prevent them from distorting the model.

We applied the interquartile range method to identify statistical outliers for each feature. Under this approach, any value that falls below Q1 minus 1.5 times the IQR or above Q3 plus 1.5 times the IQR is flagged as an outlier. The results, visualized in Figure 5, show that rainfall and temperature have the most outliers, while humidity and soil pH have none.

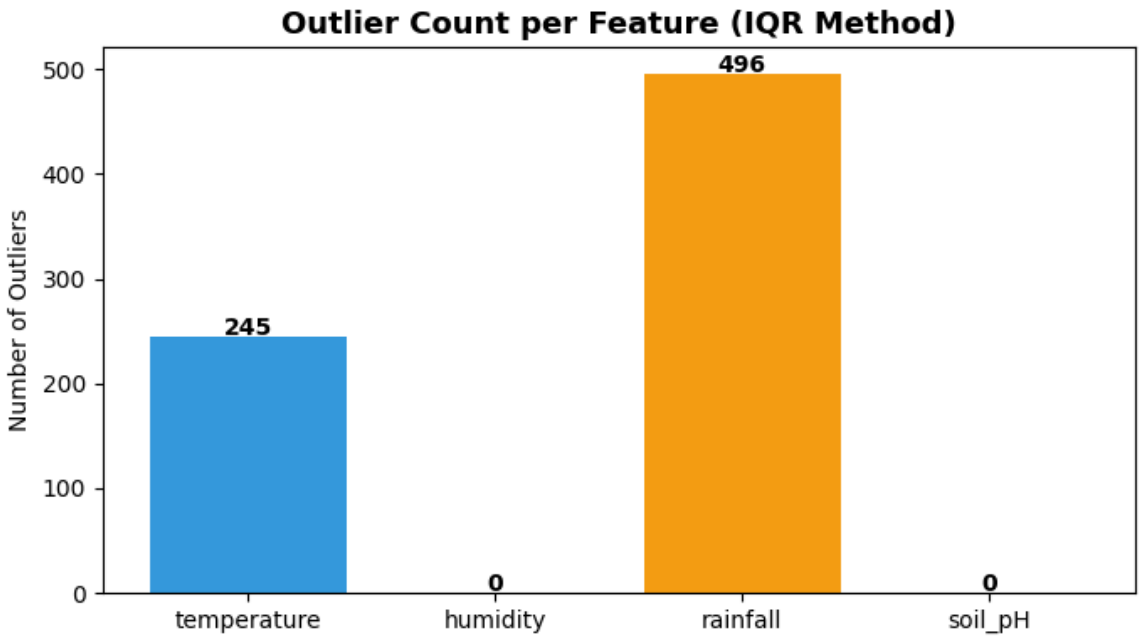

Figure 5: Number of outliers detected per feature using the IQR method

In addition to the IQR-based outliers, two domain-specific quality issues were identified. A single record has a humidity reading of 102.4%, which is physically impossible. Additionally, fifteen records have temperatures above 50°C, a threshold that is likely beyond the survival range of most crop plants and almost certainly reflects sensor or recording errors rather than genuine field conditions.

Table 5 below summarises the data quality issues identified during this analysis, along with suggested remediation strategies.

Table 5: Data quality issues identified and recommended remediation strategies

Issue Identified	Records Affected	Suggested Remedy
Humidity readings exceed 100%	1	Remove or correct the record
Temperature readings above 50°C	15	Investigate source; consider removal or capping
Rainfall upper-tail outliers (IQR)	496 (~5%)	Apply log-transform or Winsorisation before modelling
Temperature upper-tail outliers (IQR)	245 (~2.5%)	Validate against domain knowledge; handle accordingly

9. Discussion

Taken together, the findings of this exploratory analysis paint a reasonably clear picture of the dataset and the challenge it presents. The data is clean in the sense that there are no missing values, and the feature types are appropriate, but it contains a number of issues that need to be addressed before modelling can proceed responsibly.

The most important substantive finding is that humidity and rainfall are the two environmental conditions most strongly associated with plant disease in this dataset. This is ecologically plausible: high moisture levels are well known to promote the growth and spread of many plant pathogens, including fungi and certain bacteria. The relatively modest differences between the two classes, combined with a three-to-one class imbalance, suggest that the classification task will be moderately challenging. A simple threshold on any single feature will not achieve good performance, which means the model will need to leverage combinations of features.

The near-zero correlation between temperature and disease status is somewhat surprising, given that temperature is generally considered an important factor in pathogen development. It is possible that the dataset does not capture enough seasonal or geographic variation for temperature effects to emerge, or that the temperature range in the data is not wide enough to produce strong

class-level differences. Either way, temperature appears to add limited predictive value in this dataset.

Soil pH shows only a weak negative relationship with disease, suggesting that acidic soils are marginally associated with disease incidence. This might reflect the preferences of pathogens or the indirect effect of pH on nutrient availability and plant health, but the signal is too weak to be relied upon in isolation.

11. Conclusion

This report has presented a thorough exploratory analysis of the plant disease detection dataset. The data set consists of 10,000 complete records capturing temperature, humidity, rainfall, and soil pH alongside a binary disease label. The analysis revealed that the dataset is imbalanced, with healthy plants outnumbering diseased plants by approximately three to one, and that humidity and rainfall are the features most strongly associated with disease occurrence.

A number of data quality issues were identified, including a physically impossible humidity value, extreme temperature readings, and a notably skewed rainfall distribution with a substantial number of outliers. These issues will need to be resolved before the dataset can be used to train a reliable model. With appropriate preprocessing, however, the dataset appears well-suited to the development of a binary classifier for plant disease detection, and the insights gained through this analysis will directly inform that work.